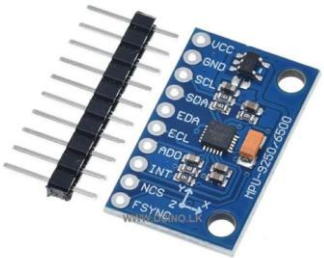
Interfacing Mobile Phone Inertial Measurement Unit (IMU) sensor data with MATLAB and Python

EE 2120: ELECTRICAL MEASUREMENTS AND INSTRUMENTATION

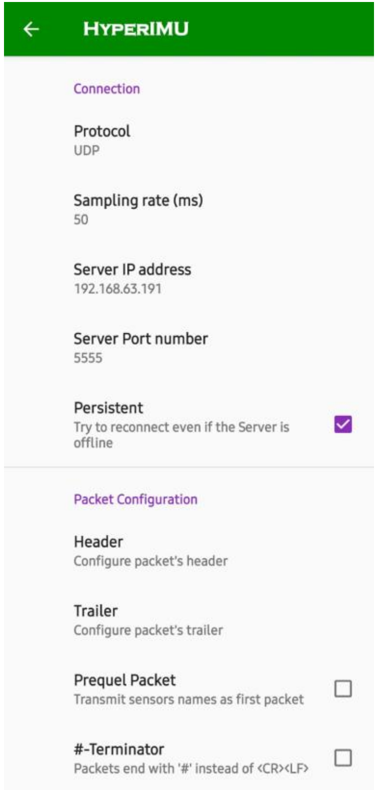
SEMESTER 4

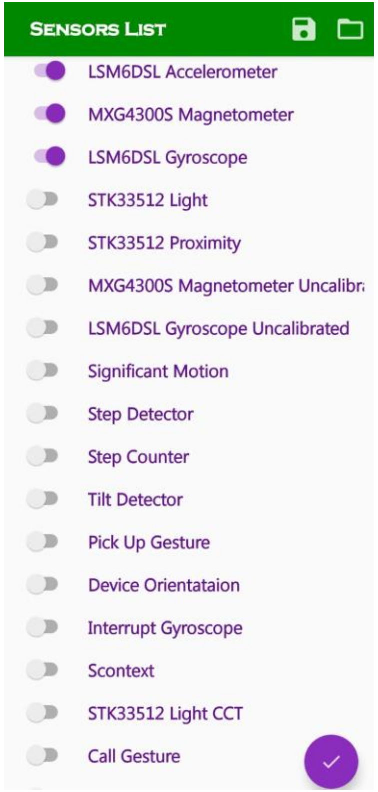
Interfacing Mobile Phone IMU with MATLAB or Python via TCP or UDP

Inertial Measurement Unit (IMU)	Attitude and Heading Reference System (AHRS)	Inertial Navigation System (INS)
▪ Raw acceleration and angular rate data from Accelerometers + Gyroscopes (sometimes Magnetometer ▪ Output: Sensor Data Only  *MPU 9250 9 DOF Accelero+Gyro+Magnetometer Module*	▪ IMU sensor + Magnetometer + Fusion Algorithms ▪ Outputs: Orientation(Roll, Pitch, Yaw)  High precision 9DOF AHRS sensor module	▪ Estimates Full Navigation state ▪ Further integrates acceleration and orientation to estimate position and velocity

Interfacing Mobile Phone IMU with MATLAB or Python via TCP or UDP

- Install Hyper IMU on Mobile Phone.
- Connect the mobile phone with the PC via either Wi-Fi or USB tethering.
  A. Wi-Fi come with delay, interference and packet loss.
  B. USB tethering needs a wired connection.
- On the PC go to command prompt and run the command, $ipconfig$ and identify the IPV4 address of the corresponding LAN Adapter.
- On the Hyper IMU app
  - Protocol: Either TCP or UDP
  - Server IP Address: IPV4 Address from previous step.
  - Server Port Number: 5555
  - Select the required sensors and Run
- In MATLAB or Python you can receive sensor data as either datagrams or bytes.
  - Port = 5555





Installing and Configuring Hyper IMU